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SuSastho.AI: A multimodal medical copilot for adolescents using evidence-based medicine and large language models

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ABSTRACT

Adolescents in Bangladesh face serious sexual, reproductive, and mental health challenges due to cultural stigma, poverty, and limited healthcare infrastructure. Within the country, 63% of adolescents are deprived of essential sexual and reproductive health services, and only 13% receive mental health support. Adolescents living in urban slums and with disabilities face additional challenges in receiving reliable health information. This limited access exposes them to a high risk of sexually transmitted infections (STIs), unintended pregnancies, and serious mental health issues. Addressing these challenges, our study introduces SuSastho.AI, a healthcare copilot providing access to reliable health information to adolescents. We utilized large language models, along with Evidence-Based Medicine, retrieval-augmented generation, and a clinically validated dataset to provide evidence-based responses, supporting both voice and text-based interactions. Clinical evaluation of a pilot study shows our method reduces incorrect responses by 26.9% and increases response correctness by 16.1% compared to other methods. It achieved an accuracy rate of 86.7% when specifically evaluated based on available knowledge. While the responses are mostly consistent with up-to-date medical practices, occasional, less precise responses highlight the need for further refinement. Participants reported overall positive feedback, where 87% found answers to their questions, and 90.7% found responses relevant. Our results show that SuSastho.AI can provide reliable and evidence-based information while being an affordable way to support traditional healthcare systems with a high potential to transform digital health. The study sets an example as an evidence-based healthcare copilot to support adolescents and lays the foundation for future research where evidence-based tools overcome social barriers.

1. Introduction

Adolescence in Bangladesh is a critical yet neglected phase of human development during which rapid biological, psychological, mental, and social changes occur. With over 20% of the population being adolescents (36 million in 2015 and rising until 2030), Bangladesh has a huge public health challenge [1]. A large gap in comprehensive sexual and reproductive health (SRH) knowledge has been linked to a high rate of child marriage, high adolescent fertility, and limited access to health services, as 63% of adolescents lack SRH services and only 13% receive mental health support [1]. Additionally, only 12%–13% of adolescents have comprehensive HIV knowledge [2]. This knowledge deficit, along with social norms and gender disparities, puts adolescents, especially girls, at high risk of sexually transmitted infections, unintended pregnancies, as well as adverse maternal and child health outcomes [3]. Along with SRH challenges, adolescent mental health is a critical yet often neglected aspect of overall well-being. Societal stigma, lack of mental health literacy, and scarcity of psychological services in Bangladesh make many experience untreated conditions like depression, anxiety, and stress-related disorders [1]. The absence of robust mental health services delays early assessment and pushes

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many to seek help only after prolonged suffering. These delays allow problems to escalate and leave young people without timely support when it matters most [4]. Moreover, the added burden of violence, abuse and drug-related challenges further magnifies the vulnerabilities of these young individuals. Adolescents living in poverty, urban slums, and other challenging circumstances face additional barriers like structural inequities and culturally ingrained practices that hinder informed decision-making [1]. The financial constraint also plays a significant role, as 79.3% of healthcare expenses are paid out of pocket [5]. At the same time, the increasing use of mobile technology, with over 61% of young individuals spending 7–8 h online daily, presents an unprecedented opportunity to bridge these gaps [6]. Altogether, these factors highlight not only the magnitude of the problems faced by Bangladeshi adolescents but also the need for innovative and scalable solutions to transform health services and outcomes for this vulnerable population.

In this fast-changing technological environment, large language models (LLMs) are revolutionizing many fields, including healthcare. Built on extensive databases, these models have transformed natural language processing (NLP) and broadened its application in many disciplines. In particular, chatbots have emerged as a versatile tool in many domains where direct human interaction is not possible nor desirable. These bots can be used in customer service, healthcare, education, productivity, and even companionship [7]. Mental health care is a prime example where AI chatbots can offer continuous, personalized support for depression, anxiety, and ADHD [8–13]. By providing accessible, stigma-free support, these bots can reduce barriers to seeking help, promote early intervention, and generate data for research and policy-making [14–17]. They use algorithms to identify patterns and generate recommendations based on individual needs and circumstances, and are capable of monitoring, communication, screening, and diagnosis. However, these opportunities also introduce challenges, such as a lack of personal interaction, dependence on technology, data accuracy, and privacy concerns, which need to be addressed [14–17]. Despite these challenges, recent surveys have found a growing number of mental health chatbot applications for various user groups [18]. While these bots are accessible and convenient, concerns about their ability to handle crises and understand user context remain unaddressed [18].

ELIZA was the first NLP computer program, a foundational step in heuristic programming and AI [19]. It used a set of rules to break down and reassemble user input to have a basic conversation by identifying keywords and limited context, proving that machines can mimic human interaction. Subsequently, AdolescentBot was the first chatbot to help Bangladeshi adolescents get sexual and reproductive health education, addressing the challenges of social taboos and lack of accessible health services [20]. The work shows that chatbots can overcome social taboos and healthcare access issues, with implications for similar public health initiatives. However, the chatbot primarily relied on limited predefined questions and answers and could not generate personalized responses. Additionally, it lacks voice-based interactions. SnehAI, a Hinglish AI chatbot, educates on sexual health and shows the potential of AI for social change [21]. It shows effective engagement and education through personalized, non-judgmental interactions and reaches vulnerable groups with significant impact. However, the chatbot relies heavily on predefined intents and rules, which may limit its ability to understand complex or ambiguous queries. Further research on chatbots in sexual and reproductive health (SRH) has shown they are a valuable tool for information and services [22]. As they can provide anonymous, non-judgmental interactions and facilitate understanding through conversational engagement, this suggests further development of chatbots to automate tasks, make conversations more authentic, and integrate them into broader service networks to better support SRH. ChatDoctor demonstrates enhanced accuracy in medical advice by fine-tuning the Llama model on real-world patient–doctor dialogues and self-directed information retrieval [23]. But the model remains vulnerable to hallucinations, where it may generate sounding but incorrect information, and no safety mechanism mitigates this risks.

In this study, we present SuSastho.AI, a novel, evidence-based, voice-activated healthcare copilot for Bangladeshi adolescents facing sexual, reproductive, and mental health (SRMH) issues. By leveraging large language models (LLMs) and Evidence-Based Medicine (EBM), SuSastho.AI helps fill the gap in accessing accurate, personalized, and culturally sensitive health information. We overcome common AI pitfalls such as hallucinations and bias by combining retrieval-augmented generation (RAG), chain-of-thought reasoning, and a clinically validated knowledge base to provide accurate answers. By supporting both text and real-time voice conversations, SuSastho.AI's multi-modal approach makes health information more accessible to adolescents, regardless of their literacy level or physical abilities. Through a pilot study, we demonstrate a significant increase in response accuracy compared to RAG and baseline LLM models. Our work outlines a replicable design framework to empower underprivileged populations and lays the groundwork for the use of AI copilots in sensitive healthcare domains.

2. Method

In recent years, large pre-trained LLMs have become the widely accepted standard in the field of natural language processing (NLP). LLMs have transformed text processing across a wide range of tasks, resulting in significant progress in both comprehension and natural language generation [24]. However, LLMs have two significant challenges: hallucination and outdated information. These are most problematic in fields like healthcare, where accurate and reliable information is crucial for addressing health questions, providing medical advice, analyzing symptoms, and offering preventive healthcare tips. It also has to provide accurate recommendations on when to see a doctor. However, LLMs can combine findings from multiple studies to derive evidence-based responses [25]. Such abilities can help in decision-making and reduce hallucinations. By synthesizing large amounts of data and extracting insights, LLMs can provide subtle interpretations that reflect the current understanding in the field. Additionally, clinical reasoning can help LLM navigate complex medical scenarios and tailor interventions to individual needs.

2.1. Evidence-Based Medicine (EBM) for adolescent SRMH

Evidence-Based Medicine (EBM) is the use of the best available research evidence to guide clinical decision-making in patient care [26]. It encourages not relying on personal opinions or conventions, but on the most objective scientific methods possible to ensure reliable results [27]. EBM can greatly improve adolescent well-being by providing reliable information and promoting healthy behaviors. We have incorporated EBM and retrieval-augmented generation (RAG) into our SuSastho.AI copilot to provide accurate, relevant, and quality health information [28]. However, EBM requires high-quality research evidence. In our case, we systematically gather and evaluate the latest medical documents, guidelines, and research on SRMH to ensure the information foundation of our copilot is up to date with current best practices. The sources are assessed for validity and reliability, including risk of bias and relevance to adolescent health care. We update and refine this knowledge base, incorporating new information as it becomes available or when an existing one is re-evaluated. This dynamic updating process is supported by our SRMH experts, who monitor emerging studies and clinical guidelines. It ensures our copilot can provide the most accurate and up-to-date information and become a reliable tool for adolescents to get SRMH information.

2.2. Response generation protocol

To ensure our copilot provides precise and relevant advice on SRMH, we have defined a protocol based on EBM. This protocol is designed to refine responses to ensure they are grounded in the best available medical information. Our response generation protocol includes the following steps:

2.2.1. Formulating a clinical question

We precisely specify the clinical question our copilot is seeking to address. It generates a focused and answerable clinical question by extracting key ideas and entities from the user's input. It ensures that the questions are specific, quantifiable, and relevant to the needs of adolescents looking for SRMH guidance.

2.2.2. Finding the best available evidence

Once we have a clear clinical question, we then find the best available medical evidence to answer it. We search our clinically validated knowledge base and select the relevant documents that match our clinical question. We use semantic similarity search using the Multilingual E5 retrieval model [29] to find the most relevant documents.

2.2.3. Assessing the evidence

Each piece of evidence found is then critically appraised to determine its validity and applicability to the clinical question and the target user group. This critical appraisal involves assessing the internal validity of the evidence, which measures the accuracy and reliability of the information for the intended population. We also assess the external validity, which measures the extent to which the information can be applied to broader populations. For example, if the information concerns adults, we carefully consider its generalizability to adolescents. We look for potential limitations, biases, or gaps in the information that could affect the answer. We also evaluate if the information is enough to provide a safe answer. Lastly, we score the evidence based on relevance for the clinical question. This scoring mechanism helps to select evidence that best aligns with the clinical question.

2.2.4. Applying the evidence

Finally, we integrate the appraised evidence into the response to provide adolescent users with accurate and tailored answers. We only use evidence with high relevance scores, so the answers are medically accurate and relevant. The system synthesizes key findings from the evidence, accounting for its strengths, limitations, and applicability to the specific clinical question.

2.3. Chain of thought (CoT) prompting

The chain of thought (CoT) prompting technique simulates human-like reasoning by decomposing complex tasks into a logical sequence of intermediate steps that progressively lead to a final answer [30]. The self-evaluation at intermediate steps helps the model arrive at a much better answer. In our case, CoT has notably improved LLM's clinical accuracy while reducing biases and hallucinations. The structured approach in CoT forces the model to follow logical steps rather than making leaps or assumptions that can lead to errors. In our work, CoT prompting is deeply integrated into the response generation protocol defined in Section 2.2. We explicitly design prompts to guide the LLM through a step-by-step reasoning process, mirroring the evidence-based clinical reasoning used by healthcare professionals. This structured reasoning is enforced using two CoT-based tasks:

2.3.1. Search query generation (Task-QG)

This task involves formulating clinical questions from user input. It uses the user's current and previous queries to extract and understand health-related concerns such as symptoms or medical conditions. Initially, it clearly defines the user query and extracts all relevant information from the conversation history. It then generates the clinical questions as search strings in both English and Bengali. Each of these steps is explicitly represented in the prompt format presented in Fig. 1, ensuring the model reasons through each phase rather than jumping straight to a search string, which may lead to hallucination and inaccurate search string generation.

2.3.2. Answer generation (Task-AG)

This task begins after retrieving medical documents using search queries generated in Task-QG. The LLM reviews medical documents and identifies relevant information, such as condition, demographics, and clinical data. It then evaluates the documents according to the evidence evaluation protocol defined in Section 2.2 and generates a relevance score. Finally, it drafts a response in English and generates the final answer in Bengali. This explicit multi-step approach is embedded in the prompt structure presented in Fig. 2, guiding the LLM to reason logically and transparently. This CoT prompting prevents the LLM from making leaps and unwanted assumptions and forces it to think logically, which leads to a much more accurate final response.

2.4. Data

Data collection and preparation were done with utmost care to ensure quality and accuracy. We combined multiple data sources and strict validation protocols to make the copilot reliable and sensitive to complex medical questions. The foundation of our training data is two-fold: clinically validated medical content and real user queries. We identified online sources that included web content, crowd-sourced content, and multimedia data, including images, audio, and video. Nevertheless, just because a resource is online and readily available does not guarantee that it is grounded by evidence [26]. For instance, Wikipedia is frequently used as a resource for addressing medical questions. However, it may contain significant gaps and has been deemed insufficient for use in Evidence-Based Medicine (EBM) [31–33]. To address this challenge, a team of clinicians verified the authenticity of the data sources.

Our Knowledge Base (KB) data are collected from authoritative sources, including guidelines from the Bangladesh Ministry of Health, materials from national NGOs, peer-reviewed literature on SRMH in South Asia, and regional documents from WHO and UNICEF. Additionally, locally produced resources and data generated through community engagement were incorporated to ensure cultural relevance. Data was collected through a combination of automated scraping and manual effort. The collected data underwent a two-layer review process by two different clinician teams, with a third team resolving any disagreements between the first two to ensure the accuracy and appropriateness of the information. Our final knowledge base (KB) contains 2063 medical documents across 20 SRMH categories, as shown in Fig. 3.

To ensure our system could accurately interpret the unique ways adolescents express health concerns, including the use of slang, idioms, indirect phrasing, and Banglish (Bengali written in English characters), we curated a task-specific dataset grounded in authentic adolescent communication. The dataset consists of 8614 training examples, including 5396 Task-QG and 3218 Task-AG instances. These examples were developed by clinical experts, drawing on insights from focused group discussions and commonly observed adolescent communication patterns. To achieve cultural and contextual relevance, our experts carefully crafted queries using colloquial language, indirect expressions, and frequently used terms, replacing standard phrasing with words and styles typical of adolescent interactions. This data was prepared iteratively by our SRMH experts, who not only ensured clinical accuracy but also paid special attention to interpreting ambiguous, colloquial, or indirect queries. We then fine-tuned our LLM on this dataset to understand and accurately extract the core clinical intent from adolescent-style queries.

2.5. System architecture

The system architecture of SuSastho.AI, presented in Fig. 4, is designed to enable seamless interaction between adolescents and the copilot. The backend service, built with the Spring Boot framework, manages response generation, stores user profiles, logs questions and

Input Example

Instruction

As a medical AI, your task is to create search strings for the user's Query. Follow these guidelines.

1. Write a simpler search string that will be used by a vector search database to get relevant pieces of information.
2. If the user mentioned any relevant information in previous messages mention that in the search string.
3. Focus on the current Query. All Previous Queries are already been answered. Previous queries were asked sequentially from top to bottom by the user. Each line contains a message.
4. Create search strings to find all relevant information of the user's Current Query.
5. Use English for Step 1 & 2.
6. If no search string is required write 'Not Required' as a search string.

Previous Queries

মেয়েদের ইস্টোজেন এর মাত্রা কমে গেলে কি কি অসুবিধা হয়
মাসে দুইবার পিরিয়ড কেনো হয়?
আমার মুখে প্রচুর পরিমাণ ব্রন তা হচ্ছে কেনো?
সাদাস্রাব কেনো হয়?

Current Query

ইদানীং অনেক ডিপ্রেশনে থাকি কেনো?

Response Format

Step 1: Understand the Query
- Clearly state the user's query, emphasizing key medical terms, conditions, treatments, concerns, or any health-related inquiry.
- Assess whether the query is broad or seeks specific information.
- Extract all relevant information about the user based on the current and previous queries.

Step 2: Generate Search String
- Write each search string in each line in English.

Step 3: Generate Search String
- Write search strings in Bengali from Step 2.

Output Examples

Step 1: Understand the Query
- The user is experiencing frequent depression and is seeking to understand the possible reasons behind it.

Step 2: Generate Search String
- Causes of Depression
- Symptoms of Depression
- Ways to Cope with Depression

Step 3: Generate Search String
- ডিপ্রেশনের কারণ
- মানসিক অবসাদের লক্ষণ
- ডিপ্রেশন মোকাবিলা করার উপায়

Fig. 1. Prompt format of the search query generation task with Input and Output example. SuSastho.AI's Large Language Model (LLM) takes the input section and generates the output section following the provided response format.

answers, and stores user voice data for model improvement. The backend has a dashboard shown in Fig. 5 to monitor the systems in real time. When the user sends a voice input, the Speech Recognition module transcribes it, the LLM module generates a response, and the Speech Synthesis module converts it into a voice response. We used Wave2Vec and an N-gram language model for Speech Recognition and the VITS model for Speech Synthesis [34–36].

The user interface shown in Fig. 6 is designed to provide an engaging and intuitive experience for adolescent users through both text and voice interaction. The app is built using the Flutter framework for cross-platform compatibility. It is available on both mobile devices and as a web application, ensuring accessibility to a wider user base.

Our medical document retrieval module is responsible for finding relevant medical information from search queries. It takes validated medical documents from the Knowledge Base (KB) and applies context-aware chunking, ensuring each document chunk contains the appropriate clinical context. This step is especially critical in a medical setting, where partial data may lead to misinterpretation or even harmful outcomes. Incomplete or out-of-context information can result in clinical errors, misdiagnosis, or inappropriate treatment recommendations. By applying this step, we ensure each chunk maintains the necessary clinical details. We then remove similar chunks using the Hamming distance, a metric that quantifies the difference between two strings. This final set of document chunks is used during the embedding search. We use the Multilingual E5 [29] embedding model for document retrieval, which performs well in Bengali. The embedding model converts each document chunk and the query into fixed-length vectors, and then computes a similarity score between the query and each document chunk to find the best match. Only documents with similarity scores above a predefined threshold are returned.

The response generation process starts with Task-QG, which generates search queries for the medical document retrieval. This task takes the current query along with a few previous queries and generates search strings. It then uses the medical document retrieval module to find the most relevant medical document chunks. If no relevant document is found, the system responds that the query cannot be answered; otherwise, it moves to Task-AG. Here, we process three sets of document chunks, ordered and selected by relevance score from highest to lowest. This is necessary because many documents in search results often exceed the LLM's context window capacity. Also, since the embedding model was not trained with a medical focus, more relevant documents may rank lower in the search results. Therefore, generating responses from multiple document sets increases the likelihood that the LLM will find the correct answer. We then evaluate the relevance score of each generated response from the internal step of Task-AG and select the response with the highest score.

We fine-tuned the Llama 3 8B Instruct model on our dataset, which serves as the core language model for both Task-QG and Task-AG. Supporting models (e.g., for speech-to-text, text-to-speech, and document embedding) facilitate input/output processing and document retrieval but do not generate content or directly influence clinical reasoning; the bulk of the system's reasoning and generation is performed by the Llama 3 8B Instruct model.

Input Example

Instruction

As a medical AI, your task is to answer users query from provided document. Strictly follow these guidelines:

1. Don't assume or extrapolate beyond the given document.
2. Focus on current Query. All Previous Queries are already been answered. Previous queries were asked sequentially from top to bottom by the user. Each line contains each message.
3. Complete the output from Step 2.
4. Strictly follow the Output Format in your response. Use English for Step 2 to 6.

Previous Queries

সাদা স্রাব কেন হয়? এর প্রতিরোধ ও প্রতিকার এ করনীয় কি?
পিরিয়ড এর সময় অসহ্য যন্ত্রণা হয়। সেটা লাঘব করার উপায় কি?
সুগন্ধ যুক্ত জয়া স্যানিটারি ন্যাপকিন ব্যাবহার করা কি নেতিবাচক প্রভাব ফেলবে?
কতো বছর বয়সে মেয়েরা বিয়ে করলে তাঁর স্বাস্থ্যের জন্য উপযুক্ত?

Current Query

পিরিয়ড হলে অনেক পেটে ব্যাথা কেনো হয়?

Document

FAQ: আর কী কী কারণে পেট ব্যথা হয়?
Answer: পেট ব্যথার ধরন সম্ভাব্য কারণ - পেটের নীচের অংশে ডান দিকে হঠাৎ ব্যথা শুরু হওয়া অ্যাপেন্ডিসাইটিস ....

FAQ: যোনিপথে ব্যথার কারণ কী?
Answer: * পেটে (পেটের অঞ্চল) ট্রমা
....

Response Format

Step 1: Understand the Query
- Clearly state the user's query, emphasizing key medical terms, conditions, treatments, concerns, or any health-related inquiry. Do not assume anything from the document.
- Assess whether the query is broad or seeks specific information.
- Extract all relevant information about the user based on the current and previous queries.

Step 2: Analyze the Documents
- Retrieve all relevant information from the provided document. This could include demographic information, clinical data, statistical data etc.

Step 3: Specific Relevance & General Applicability
- Identify and describe specific parts of the provided context that directly address and support the answer to the query.
- Determine how well the document's content applies to the specific conditions, populations, or scenarios mentioned by the user.
- Consider factors such as the age group, gender, disease specificity, and geographic relevance.
- Evaluate the documents broader relevance and applicability to the wider audience or more general conditions related to the query. If relevant, describe how.

Step 4: Answer Analysis
- Based on the analysis in Steps 1, 2 and 3, how closely does the document address the user's query?
- What are the potential limitations, biases, or gaps in the documents that could hinder answering the medical query accurately?
- How might these limitations affect the overall reliability and comprehensiveness of the answer?
- Is the information enough to provide a meaningful and safe answer?

Step 5: Relevance Score
- Only write the relevance score of the document for the user query based on analysis in Step 3 & 4. Score range should be 0 to 5.

Step 6: Final Answer
- Based on the analysis in Step 4, present the answer in a clear and understandable format to the user. Never mention about the document in your answer as it is hidden from the user. If the query can't be answered, you must say "I don't have enough information" and provide a general advice without adding extra information.

Step 7: Final Answer
- Rewrite the final answer based on Step 6 in Bengali. Do not perform direct translation. The answer must be natural.

Output From Task-1

Step 1: Understand the Query
- The user is asking about the reasons behind experiencing significant abdominal pain during menstruation. This query seeks specific medical information related to menstrual cramps and their underlying causes.

Output Example

Step 2: Analyze the Documents
- The document provides information on various causes of abdominal pain, including menstrual pain. It specifically mentions that menstrual pain or cramping is a common occurrence during menstruation.

Step 3: Specific Relevance & General Applicability
- The document directly addresses the user's query by stating that menstrual pain is a common symptom during menstruation. This information is specifically relevant as it confirms that experiencing abdominal pain during periods is a recognized medical condition.
- The information is applicable to the general population of menstruating individuals, making it broadly relevant. It does not specify any particular age group or demographic, which implies its general applicability to all individuals experiencing menstruation.

Step 4: Answer Analysis
- The document closely addresses the user's query by confirming that abdominal pain during menstruation is a common symptom, known as menstrual cramps.
- There are no significant limitations or biases in the document regarding this specific query. However, the document does not delve into detailed causes or mechanisms behind menstrual cramps, which could be seen as a gap if the user seeks a deeper understanding of the physiological processes involved.
- The information provided is sufficient to affirm the normalcy of menstrual cramps but lacks depth for those seeking more detailed explanations or management strategies.

Step 5: Relevance Score
- 4

Step 6: Final Answer
- Menstrual pain or cramping during periods is a common occurrence and is recognized as a normal part of the menstrual cycle.

Step 7: Final Answer
- মাসিকের সময় পেটে ব্যথা বা কামড়ানো একটি সাধারণ ঘটনা এবং এটি মাসিক চক্রের একটি স্বাভাবিক অংশ হিসেবে গণ্য করা হয়।

Fig. 2. Prompt format of the answer generation task with Input and Output example.

2.6. Model evolution and selection

In the course of our research, we initially experimented with RASA and Chatterbot [37,38]. However, these chatbot frameworks did not meet our requirements for handling comprehensive question-answering tasks. Subsequently, we explored the Bloom model for its proficiency in Bengali [39]. However, Bloom's primary limitation was its focus solely on text generation, and it lacked a conversational variant. The sub-optimal results in the question-answering task and occasional harmful answer generation made it unusable for a medical context. Implementing retrieval-augmented generation (RAG) [28] in Bloom improved the quality of the answers and reduced both hallucinations and harmful answer generation, but the issue was not resolved completely. Although new problems emerged with context recognition, for instance, the model would assume the user is pregnant when answering unrelated questions that mention symptoms like vomiting. We resolved this by implementing a keyword-based filter that removes pregnancy data from the context documents when the user query is not related to pregnancy. However, in certain situations, some pregnancy-related data may apply to a wider population. The problem was resolved by incorporating EBM guidelines, so the keyword-based filter was no longer required. After adopting the Llama 3 model, its enhanced Bengali comprehension significantly improved our system's ability to answer questions in both Bengali and English.

2.7. Evaluation method

Evaluating LLMs is challenging due to a lack of standard evaluation methods, especially in the medical domain [40]. Metrics like BLEU,

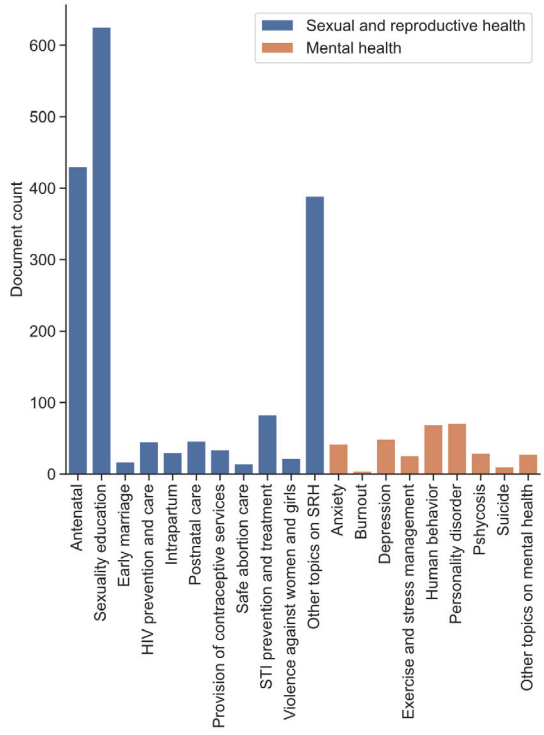


Fig. 3. Data distribution across various SRMH topics of our copilot's Knowledge Base.

ROUGE, METEOR, and perplexity are commonly used to evaluate generated text [41–44]. However, they are incapable of capturing underlying meaning, semantic nuances, relevant context, and conversational dependency [45]. They also lack medical understanding, which makes them unusable in a medical context. So human evaluation is crucial in assessing the reliability and accuracy of medical copilots [46,47]. Moreover, human insights are necessary to evaluate cultural relevance, complex user interaction, and ambiguous input-handling capability. For this reason, we primarily relied on human evaluation, a process that proved to be extensive and labor-intensive. Our SRMH experts evaluated the responses for accuracy, factual consistency, and relevance using a checklist to ensure consistent assessment. Two independent evaluators were involved in the process, and a third was consulted in case of disagreement. We conducted a pilot study to evaluate and compare the system's performance. To ensure safety and ethical standards, adolescents were not included in this initial trial. The study involved adolescent health professionals and quality assurance experts to evaluate and identify areas for improvement before broader testing with the intended population.

We evaluated SuSastho.AI based on the available information in its Knowledge Base (KB). In this evaluation, we expected the copilot to provide correct answers when the information was present in the KB and to appropriately refuse to answer when it was not. If the copilot correctly refused to answer due to the absence of information in the KB, we considered it a correct response. Conversely, if it failed to provide a correct answer or refused to provide an answer when the information was in the KB, we counted it as incorrect. The question–answer pairs used in this evaluation were sourced from the pilot study.

To assess the effectiveness of our method, we compared SuSastho.AI with the Retrieval Augmented Generation (RAG) method and a baseline Llama 3 8B model with the following configuration:

- RAG + Llama 3 8B + Validated KB: This setup uses our validated KB for document retrieval and then concatenates the retrieved passages with the user query, which are provided as input to a Llama 3 8B instruct model. The model is only responsible for generating answers based on the combined input, without any additional reasoning or structured protocol.
- Llama 3 8B: Here, the Llama 3 8B instruct model is directly given the user queries without access to any external information or KB.

In both experiments, we used the default pretrained weights of the Llama 3 model, without any fine-tuning. LLMs were instructed to provide concise answers in Bengali for adolescents. This evaluation focused on the correctness of the generated answers, using the same questions as in the pilot study.

3. Results

3.1. Pilot study outcomes

A total of 32 participants attended the pilot study. Among them, 81.3% were male, and 18.8% were female, and ages ranged from 23 to 55 years (mean: 29.5, SD: 5.9 years). Prior to their participation, all participants were briefed on the study's purpose and limitations, as well as the nature of the interaction with the copilot. Informed consent was obtained from each participant, ensuring adherence to ethical research standards. By the end of the study, participants had asked a total of 428 questions. Some of the most commonly used words in the questions are presented in Fig. 7. Question length ranges from 2 words to 38 words. There were 9 greetings consisting of a single word. Answers range from 5 words to 113 words.

When the system refuses to answer, it responds with a fixed-length message of 9 words politely refusing to answer. One answer with 113 words repeated specific tokens to the end. This problem was not present in any other answer. The other long answers were mostly correct. The relationship between question and answer lengths and their distributions is presented in Fig. 8.

Fig. 9 visualizes the relationship between different categories of questions and answers. For questions that were explicitly addressed by the knowledge base (n=199, 46.5%), 162(81.4%) were based on explicit knowledge base (KB) information, 22(11.1%) were based on implicit KB information, 7(3.5%) contained fictitious information, and 8(4%) questions the system refused to answer. When the question was indirectly or partially covered by the KB(n = 75, 17.52%), 41(54.7%) answers were based on explicit KB information, 14(18.7%) were implicit KB-based answers, 17 (22.6%) had fictitious information, and 3(4%) questions the system refused to answer. For questions not covered by KB (n = 121, 28.3%), 11(9.1%) answers were based on explicit KB information, 5(4.1%) were implicit KB information, 25(20.7%) had fictitious information, and 80(66.1%) questions the system refused to answer. In situations where participants provided statements or greetings (n = 33, 7.7%), 1(3%) answer had implicit KB information, 8(24.2%) answers had fictitious information, and 24(72.7%) questions system refused to answer. Table 1 presents the evaluation results, conducted based on the information available in the KB. It details the individual accuracy for each question and answer category. Overall, SuSastho.AI provided answers that were 86.68% correct, 2.57% partially correct, and 10.75% incorrect.

3.2. Error categorization and performance analysis

We further analyzed SuSastho.AI's responses to identify the sources of the errors. For questions explicitly covered by the KB and whose answers contain explicit KB information, the wrong answers are primarily due to irrelevant information in the answers. Answers with implicit KB information: only three answers were incorrect out of 22 answers, due to the wrong information provided in the answer. For fictitious answers, there were two wrong answers due to ambiguous questions. All refused answers were considered incorrect, as the LLM should respond since the questions were explicitly in the KB. For questions indirectly/partially

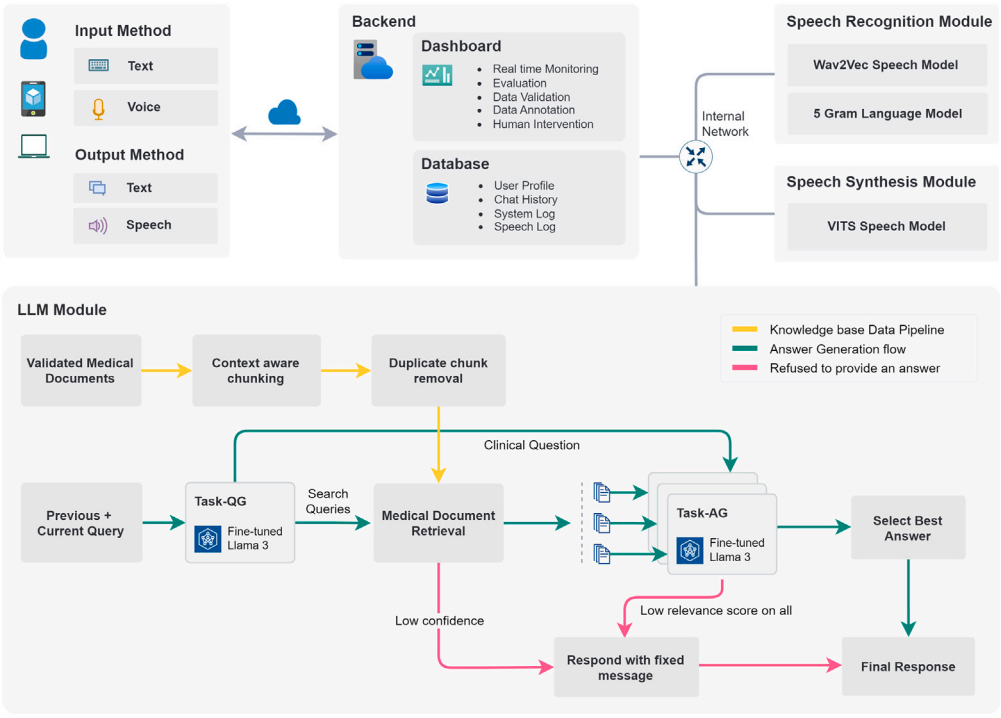


Fig. 4. Overview of the SuSastho.AI system architecture, illustrating the interaction flow between the adolescent user and the copilot. The architecture consists of a user-friendly mobile app for input, a backend service for processing, and integrated modules for speech recognition, fine-tuned Llama 3 8b model-based response generation, medical document retrieval, and speech synthesis. The backend manages user data, logging, and real-time monitoring through a dedicated dashboard, ensuring seamless and reliable medical query processing.

Table 1
Performance Metrics of SuSastho.AI Across Various Question and Answer Categories. Accuracy values are in percentage. KB: Knowledge Base, KBi: Knowledge Base Information, info: Information.

Type		Answer Accuracy (%)		
Question Type	Answer Type	Correct	Partially Correct	Incorrect
Question explicitly covered by KB	Answer based on explicit KBi	91.3	6.8	1.9
	Answer based on implicit KBi	86.4	0	13.6
	Answer has fictitious info.	71.4	0	28.6
	System Refused to answer	0	0	100
Question indirectly covered by KB	Answer based on explicit KBi	78.1	14.6	7.3
	Answer based on implicit KBi	92.9	0	7.1
	Answer has fictitious info	88.2	5.9	5.9
	System Refused to answer	66.7	0	33.3
Question not covered by KB	Answer based on explicit KBi	81.8	18.2	0
	Answer based on implicit KBi	80	20	0
	Answer has fictitious info	68	4	28
	System Refused to answer	100	0	0
Statement or greetings	Answer based on explicit KBi	–	–	–
	Answer based on implicit KBi	0	0	100
	Answer has fictitious info	37.5	0	62.5
	System Refused to answer	100	0	0
Total		86.68	2.57	10.75

covered by KB and the answers that have explicit KB information, wrong answers occurred due to the LLM not understanding the questions, or the information in the answer is not relevant to the question. For the fictitious answers, one answer was not relevant to the question, and another contained an incorrect fact. Among the refused answers, two questions lacked the required information in the KB, so we considered them correct; the other answer was wrong because it had information in the KB, but the LLM could not use. For questions not covered by KB but with answers that had explicit KB information, 10 answers were correct, and only one was partially correct. For answers containing fictitious information, seven questions were incorrect out of 25 questions. The incorrectness is due to the question being ambiguous or the LLM misinterpreting it. In the correct answers, LLM suggested a general solution. In the remaining 80 questions, the copilot refused to answer; this was the intended behavior, so we considered all as correct. When a statement of greetings is provided, the model should also refuse to answer, as we decided to limit the copilot to answering only questions to ensure safety. However, for one question, it gave a generic answer using the implicit KB information, and for 8 questions, it gave fictitious information. Out of the 8 questions, 7 were asked in Banglish.



Fig. 5. SuSastho.AI backend dashboard for real-time monitoring, response review, and system health tracking to ensure quality and reliability.



Fig. 6. SuSastho.AI mobile app, designed for adolescents with an intuitive interface supporting voice and text queries for accessible medical information.



Fig. 7. Word Cloud displays English translations of key terms in questions, visualizing the topics of greatest interest to the users of the Pilot Study. Font size represents the frequency of the word.

3.3. Comparative analysis

Fig. 10 shows the comparison of response quality among different methods. We used the same 428 questions from the pilot study to compare SuSastho.AI with RAG and the baseline Llama 3 8B model. SuSastho.AI performed better, with 61.9% correctness, much higher than RAG (40.4%) and Llama 3 (45.8%). Moreover, SuSastho.AI had a lower no-answer rate (26.9%) than RAG (42.3%) and a higher rate than Llama 3 (1.6%). It had fewer partially correct answers (2.6%) than RAG (4.4%) and Llama 3 (17.1%). SuSastho.AI also had lower incorrect answers (8.6%) than RAG (12.9%) and Llama 3 (35.5%). Compared to the RAG and baseline Llama 3 8B model, our copilot reduced incorrect responses by 4.3% and 26.9%, respectively, and increased correct responses by 21.5% and 16.1%, respectively. This

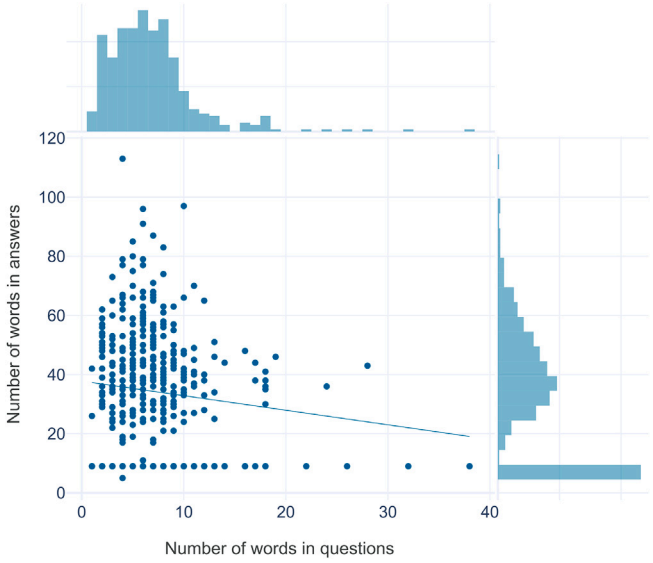


Fig. 8. Scatter plot visualizing the number of words in questions (X-axis) and number of words in answers (Y-axis) generated by SuSastho.AI during the pilot study. Trendline is plotted using the Ordinary least squares regression method. Marginal histograms are included to show the distribution of word counts.

evidence shows our copilot is more reliable and accurate in providing health information and is a more reliable resource than other methods.

3.4. Qualitative analysis

We conducted a qualitative evaluation of the methods to evaluate the effectiveness of the responses. Based on the evaluation, SuSastho.AI demonstrates superior performance. It provides more accurate, detailed, and safer responses, particularly crucial for numerous health-related queries. It also handles sensitive topics responsibly and consistently refuses inappropriate or unanswerable questions. The RAG + Llama 3 method refuses more answers, and also the response quality is low. The answers provided are often generic and sometimes irrelevant or inaccurate. This system seems significantly less capable and overly constrained compared to the others. The baseline Llama 3 model responses are more conversational but suffer from low accuracy and potentially misleading information. It also attempts to answer a wider range of questions with a tone that is often conversational and empathetic. Moreover, it provides reasonable general advice for some common questions. However, the model occasionally reinforces stereotypes related to gender and health, which was not observed in other methods. Table 2 presents responses to two questions. In this example, our model provided well-structured and informed answers. It produced detailed, medically relevant responses. On the other hand, RAG + Llama 3 responses were repetitive or uncertain, and baseline Llama 3 responses were often vague or shallow. These results suggest that our system is more suitable for deployment in health information settings where trust, clarity, and accuracy are critical.

3.5. Usability questionnaire

We developed a custom usability questionnaire to assess context-specific factors, including clinical usefulness, understanding of adolescent phrasing, tone, and cultural appropriateness. Our clinical experts designed the questionnaire to be linguistically and culturally relevant to adolescents while still including core usability items, such as ease of interaction, satisfaction, and response time. The results, shown in Fig. 11, indicate user sentiment towards SuSastho.AI. Among the users, 53.1% agreed, and 34.4% strongly agreed that they found the answers

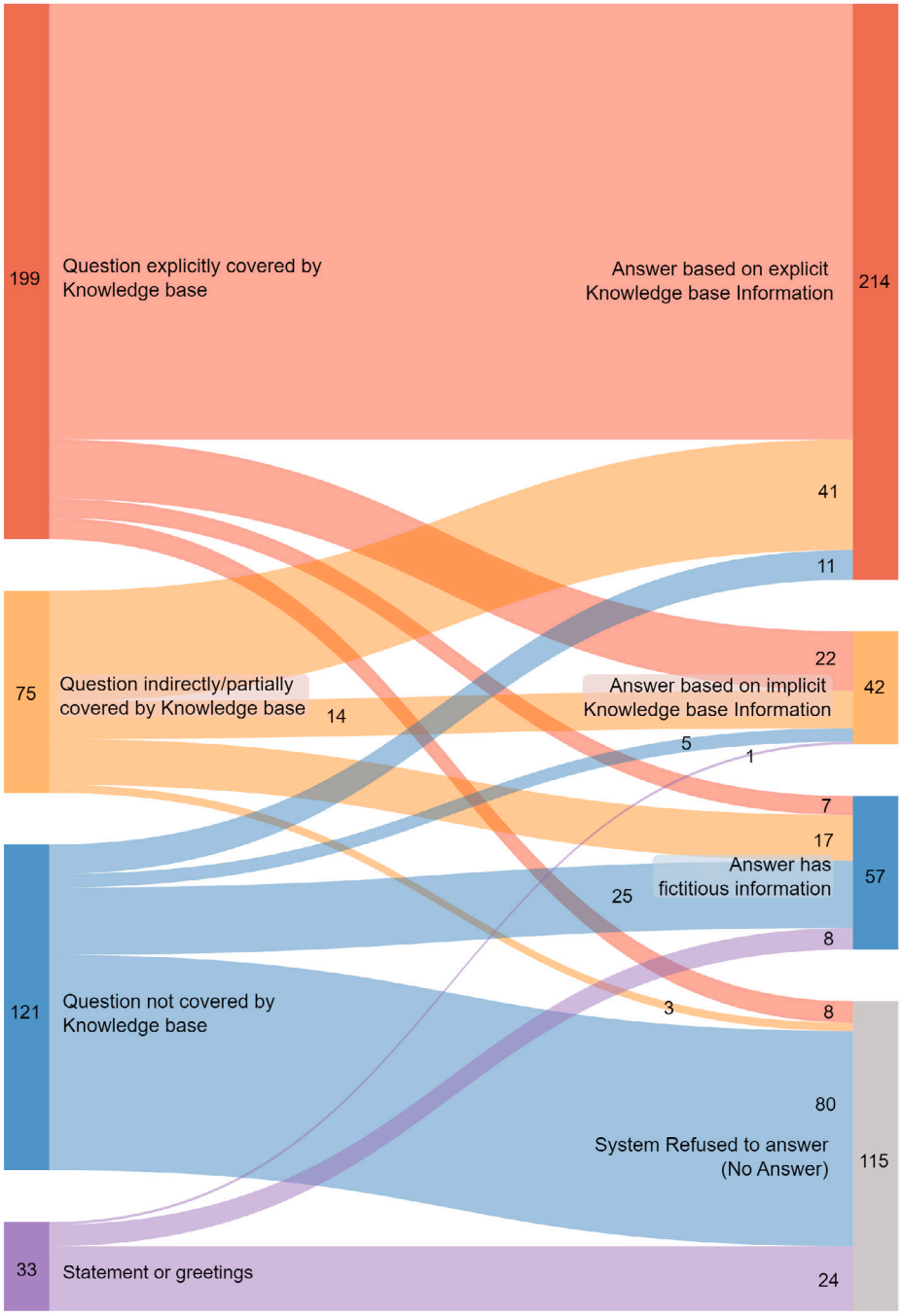


Fig. 9. Visualization of the relation between categories of questions and SuSastho.AI's answers. Question types are on the left, and answer types are on the right. The numbers represent the total question–answer pairs in each category or connection.

they were looking for. High agreement on relevance (71.9% agreed, 18.8% strongly agreed) and usefulness (65.6% agreed, 31.3% strongly agreed) of answers. Response time was liked by most of the users (71.9% agreed, 9.4% strongly agreed, and 3.1% strongly disagreed). Overall satisfaction with the copilot experience was: 62.5% agreed, and 21.9% strongly agreed. Many found the answers informative, but 25% disagreed that the answers felt human, so there is a perception of artificiality in the interaction. Disagreement was also high on the copilot's understanding of the question (21.9%).

4. Discussion

In this study, we presented methods to overcome the constraints of LLMs in clinical question answering by efficiently combining EBM and RAG with up-to-date evidence to generate precise answers. We also showed that EBM, when combined with a chain of thought, can produce medically accurate responses. Our copilot's ability to provide clinically accurate answers demonstrates its improved understanding of medical concepts, which is crucial when accuracy affects patient outcomes. It worked well within well-defined and limited medical scopes covered in the knowledge base (KB). However, it struggled with less common medical conditions and issues that are not well represented in the KB, therefore emphasizing the need for ongoing KB updates. The copilot also struggled with ambiguous and Banglish (Bengali written in English characters) input, which calls for more work on dataset improvements and model training. Our evaluation shows that SuSastho.AI performs

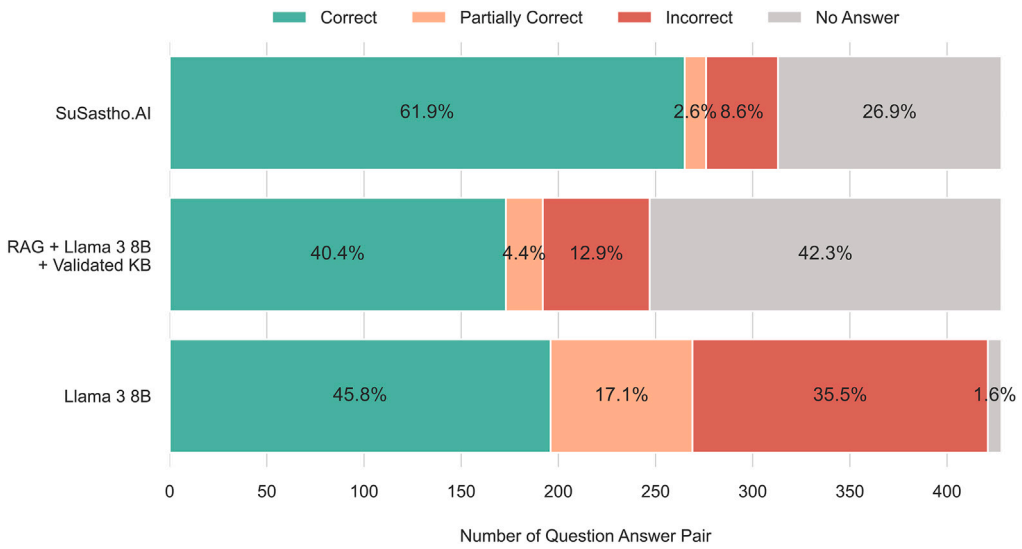


Fig. 10. Comparison of response accuracy across SuSastho.AI, Llama 3 model, and Llama 3 with Retrieval Augmentation Generation (RAG) method; KB: Knowledge Base.

Table 2
Sample responses of each method (translated from Bengali).

Question	Method	Output
What is postpartum depression and how to deal with it?	Ours	Postpartum depression is a condition where new mothers may feel sad, empty, and more emotional after giving birth. These feelings usually last for a few days, but if these symptoms last for more than two weeks, it could be a sign of postpartum depression. Hormonal changes after giving birth can also cause this condition. If symptoms persist or worsen, it is important to seek treatment.
	RAG + Llama 3 + Valid. KB	Sorry! Postpartum depression is a type of mental health problem that occurs in new mothers. It can occur after giving birth and is a type of mental health problem in the mother. It is a type of mental health problem in the mother that can cause her to suffer from mental illness. It can be dealt with if the mother's mental health is regularly checked and her mental health problems are discussed with the doctor.
	Llama 3	I want to help you! Postpartum depression is a common condition in which a mother feels depressed after giving birth. It usually lasts for 1-3 weeks after delivery. To cope with it, a mother can do her part by addressing her health issues, getting support, and seeking social support.
Why does my chest hurt?	Ours	Chest pain can be caused by a variety of things, including heart problems, gastroesophageal reflux disease, excessive anxiety, muscle tension, and skin pain. If you have chest pain, look for signs that the pain increases with physical exertion and decreases with rest and that you have other symptoms such as shortness of breath, nausea, sweating a lot, or sudden onset of pain that radiates to the left arm. If you think you are having a heart attack, see a doctor immediately.
	RAG + Llama 3 + Valid. KB	Sorry! I couldn't find an answer to this question. But I can tell you that chest pain can be a symptom of many different conditions. It could be a heart attack, angina, gastric disease, or a symptom of some other condition. You should see a doctor to determine the cause of your chest pain.
	Llama 3	My chest pain could be due to allergies, asthma, or some other problem. You should talk to a doctor, they can diagnose your problem.

better at providing health information than other methods. Its reasoning framework and self-validation process are the key to its higher correctness rate. Noteworthy is SuSastho.AI's ability to safely refuse answers and its lower frequency of incorrect answers, making it suitable for real-world applications where accuracy is critical. Also, our copilot provides more definitive and clear information, which is important in healthcare, where any ambiguity can lead to misinterpretation and health risks. These results suggest that SuSastho.AI can be integrated into a larger health information platform to improve user experience and outcomes.

4.1. Limitations

We rely on human evaluation to measure the accuracy and effectiveness of our copilot. However, it required significant time and effort, so we could only test a limited number of samples during the evaluation phase. Hence, the evaluation was limited, and we could not test the copilot over multiple scenarios and topics. This limitation might have affected the accuracy and comprehensiveness of our evaluation, as the smaller sample size may not have reflected the copilot's performance in more varied or real-world interactions. We had multiple independent evaluators in the evaluation process to reduce personal bias, but the small number of evaluators could still introduce some bias. The evaluation process has entirely relied on the knowledge of our SRMH experts. Furthermore, additional effort is needed to thoroughly evaluate the copilot's performance, encompassing aspects such as missing content, potential harm, bias, cultural relevance, and empathy. Additionally, our error analysis revealed key challenges in building a reliable clinical chatbot. Even when the knowledge base (KB) had clear answers, the model sometimes produced irrelevant or fabricated information,

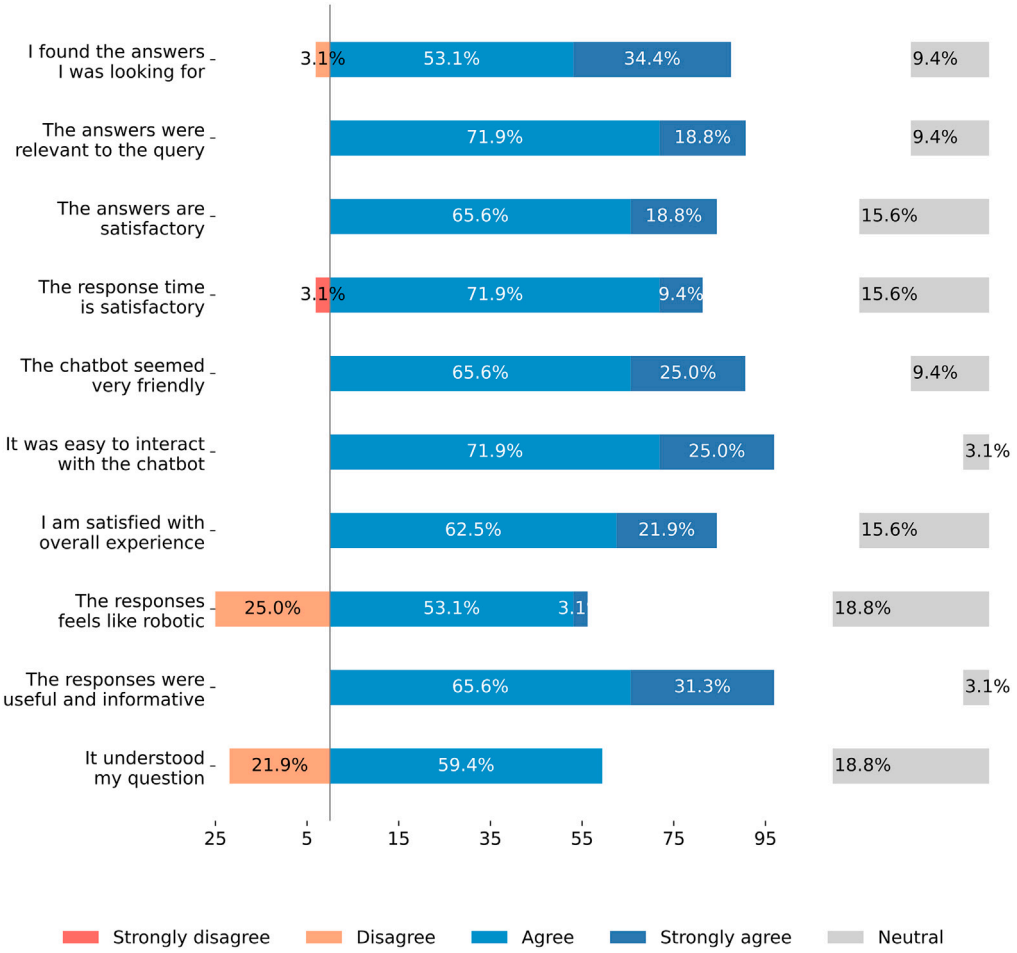


Fig. 11. User feedback on copilot's performance with the percentage of users on the X axis and feedback questions on the Y axis.

indicating issues with retrieval accuracy and contextual understanding. Banglish queries often yielded poor responses, likely due to limited mixed-language training data, highlighting the need for stronger multilingual data augmentation. While refusal behavior generally met safety requirements, we observed some inappropriate responses to greetings, suggesting the refusal mechanism could be improved. While we used CoT in the research, we have not fully explored its potential. Additional prompt engineering and exploration are necessary in this area. We also have not explored the full potential of EBM. Additional EBM guidelines can be incorporated to improve the LLM's response. The reasoning steps used in the tasks can also be optimized to improve performance. Due to limited resources, many of our research decisions were influenced by the copilot's operational costs. For instance, our current system using the Llama 3 8B model costs $0.2 per million tokens generated, which is substantially lower than $30 per million tokens for GPT-4, 150 times more expensive. Since we want to deploy the copilot widely, sustain it for a long time, and make it free for all adolescents to use, the cost of GPT-4 and similar large models makes it impractical for our usage.

Another important limitation is the linguistic and cultural gap between adolescents' everyday language and the formal language used in evidence-based medical sources. Adolescents often use unique idioms, slang, indirect expressions, and mixed languages like Banglish, which are not always well-represented in our current knowledge base or training data. This mismatch can make it difficult for the system to accurately interpret and extract the intent behind user queries, especially when questions are phrased indirectly or use culturally specific references. While our error analysis identified Banglish and ambiguous input as particular challenges, broader issues of cultural and linguistic diversity require further attention. Our usability assessment relied on a custom instrument tailored to this context, which may limit direct comparability with other studies. Future work will explore complementing it with broader measures once adapted and validated for this population.

4.2. Ethical considerations

Building a medical copilot, especially one for adolescents, raises many ethical considerations. Data protection is key to keeping user trust and complying with the law. Another concern is the copilot providing the wrong medical advice. While the system is designed to provide accurate health information, clear disclaimers are needed to inform users that the copilot's advice can be incorrect and should not replace professional medical advice. We also implemented mechanisms to block answers when copilot was not confident, and we informed users to seek advice from healthcare professionals. We made efforts to neutralize bias during data preparation and training. However, ongoing monitoring is necessary to detect and correct any bias that may arise as user interactions evolve and more data become available.

4.3. Societal impact

Adolescent health in Bangladesh is being challenged by rapid demographic changes, social norms, and service delivery gaps, so innovative interventions are not just desirable but necessary. With over a fifth of Bangladesh's population being adolescents, many of whom face barriers such as limited SRMH knowledge, high rates of child marriage,

adolescent fertility, and gender-based violence, traditional health systems have struggled to provide comprehensive and timely care [1]. SuSastho.AI addresses the limitations of current adolescent health interventions, which are hindered by misinformation, stigma, and traditional practices rooted in social taboos. Our copilot can provide accurate and context-specific responses to bridge the long-standing gap. Moreover, by democratizing access to validated health data, SuSastho.AI can empower adolescents to make informed decisions, thereby reducing risky behaviors and counteracting the impacts of social discrimination and structural poverty. This digital tool will complement government initiatives, reinforce structures such as Adolescent Friendly Health Services (AFHS) and school health programs, and extend a lifeline to underserved communities, including street children, adolescents in slum dwellings, and those in refugee status. The promising pilot study results, with a significant reduction in incorrect responses and marked improvement in the accuracy of health advice, demonstrate that SuSastho.AI can transform how adolescent health is addressed in Bangladesh.

4.4. Future work

In the future, we plan to extend its capabilities to support newborn and elderly care. We will also enrich the training data with culturally sensitive interactions to make the copilot's responses relevant and inclusive. The expansion will also address errors and biases, reducing them to a level that enables safe deployment to a wide user base. We also plan to expand our dataset with additional real-world adolescent queries, collaborate with adolescent users during evaluation and design, and explore NLP techniques to better handle idiomatic and indirect language. A critical addition in our next phase is the implementation of a risk detection system. This system will identify users at risk and connect them with healthcare professionals immediately, enabling timely and appropriate interventions. We plan to add a symptom checker to identify potential diseases, making it a preliminary diagnostic tool [48]. We will also embed Clinical Standard Operating Procedures (SOPs) within the copilot to guide its actions across different clinical scenarios, in line with established medical protocols. To improve the quality of the copilot's responses, we will explore advanced techniques such as Reinforcement Learning from Human Feedback (RLHF) [49], Direct Preference Optimization (DPO) [50], and the deployment of larger, more capable models.

5. Conclusion

In conclusion, SuSastho.AI represents a significant leap forward in using large language models (LLMs) to support the sexual, reproductive, and mental health of Bengali adolescents. It bridges the gap between health education and AI technology in an accessible, culturally sensitive way. As Evidence-Based Medicine (EBM) principles are applied, the information provided by SuSastho.AI is not only accurate but also relevant and reliable to the problems adolescents face due to social norms and limited resources. As per the pilot study and user feedback, the copilot has the potential to improve health literacy among Bengali-speaking adolescents. Although the copilot has higher accuracy and ease of use, the language subtleties and response accuracy can be improved. SuSastho.AI will be expanded in the future to include more diagnostic capabilities and health topics, making it a complete health resource. This research also shows how to create and deploy digital tools in low-resource settings and advances the field of AI in health. Our research shows how LLMs can overcome barriers to accessing reliable health information and can be a model for similar initiatives globally. As AI advances, its use in health interventions will only grow, providing more opportunities to improve public health in many contexts.

CRediT authorship contribution statement

Shamim Ahamed: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Moinul H. Chowdhury:** Writing – review & editing, Methodology. **Marzia Zaman:** Writing – review & editing, Validation, Investigation, Data curation. **Abhishek Agarwala:** Writing – review & editing, Data curation. **Md Abdullah Al Mashud:** Writing – review & editing, Software, Investigation. **Tariqul:** Writing – review & editing, Validation, Resources, Project administration, Methodology, Investigation. **Tareq Al Muntasir:** Writing – review & editing, Methodology. **Rifat Rahman:** Writing – review & editing, Methodology. **Rifat Shahriyar:** Writing – review & editing, Methodology. **Farhana Sarker:** Writing – review & editing, Supervision, Resources. **Khondaker A. Mamun:** Writing – review & editing, Supervision, Resources, Validation, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Code availability

All code, fine-tuned model weights, prompt templates, and knowledge base data, along with step-by-step instructions to ensure replicability, are publicly available at: https://github.com/BanglaGPT/SuSastho.AI

Ethics statement

This research strictly adheres to ethical standards and local laws. It has been approved by the Bangladesh Medical Research Council (BMRC), the national body responsible for coordinating and promoting health and medical research in Bangladesh, under Reference No.: BMRC/ERC/RPF-Bill/2024-25/1279 on November 26, 2024. In addition, it has received ethical approval from the Institutional Research Ethics Board (IREB) of United International University, under Reference No.: IREB/2023/019 on August 16, 2023. All participants provided informed consent before taking part in the study. Their privacy and confidentiality were strictly maintained, and data were anonymized to ensure compliance with ethical standards.

Declaration of generative AI in scientific writing

During the preparation of this work, the author(s) used 'Grammarly' in order to improve the readability and language. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The dataset created and used in this study was not deposited publicly due to incorporating proprietary data. However, all data are available upon request from the corresponding authors for replicability.

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